Github Repo

1 Introduction and Background

X-ray scattering produces a 1-D intensity profile $I(q)$ that encodes the 3-D shape of a molecule, but computing it exactly via the Debye equation scales as $O(N^2)$ in atom count, making it prohibitively slow for large structures and incompatible with gradient-based optimization. This proposal seeks to answer the question *can a graph neural network learn to predict $I(q)$ directly from atomic coordinates?* To answer this, this project aims to train a message-passing GNN as a fast, differentiable surrogate for the exact computation.

Graph neural networks for molecular property prediction were pioneered by SchNet [1], which introduced continuous-filter convolutions on interatomic distances and remains a strong baseline for scalar molecular properties. DimeNet [2] extended this by incorporating bond angles, improving accuracy on quantum-chemical targets such as the QM9 benchmark [3]. On the scattering side, CRYSOL [4] established the spherical-harmonic decomposition approach that we use to generate ground-truth $I(q)$ labels. More recently, Schütt et al. [5] introduced PaiNN, an equivariant message-passing network.

When a beam of X-rays is fired at a molecule, each atom deflects ("scatters") the beam by an amount that depends on the atom's identity and the deflection angle q. Each atom has a characteristic *atomic form factor* $f_k(q)$ encoding how strongly it scatters at angle q. By measuring the total scattered intensity across all angles, we obtain a *scattering profile* $I(q)$ which is a unique fingerprint of the molecule's 3-D shape [6]. The exact formula, the **Debye equation**, sums the pairwise interatomic interactions:

$$I(q) = \sum_{\alpha} \sum_{\beta} f_\alpha(q)\, f_\beta^*(q) \;\operatorname{sinc}\!\big(q\,|\mathbf{r}_\alpha - \mathbf{r}_\beta|\big) \tag{1}$$

where $\operatorname{sinc}(x) = \sin(x)/x$ captures the intuition that atoms far apart interfere less strongly. Because every pair must be evaluated, this scales as $O(N^2)$ in the number of atoms. The Stuhrmann decomposition (Eq. 2) expands the Debye equation into a sum of spherical harmonics, decomposing the molecular shape into a sum of L^2 angular modes, truncated at order L. This reduces the time complexity down to $O(N \cdot L^2)$, and assuming for most cases $L \ll N$, reduces to $O(N)$ [7]. This project uses the in-vacuo form (no solvent, hydration shell, or intermolecular forces); a live demo is available online.

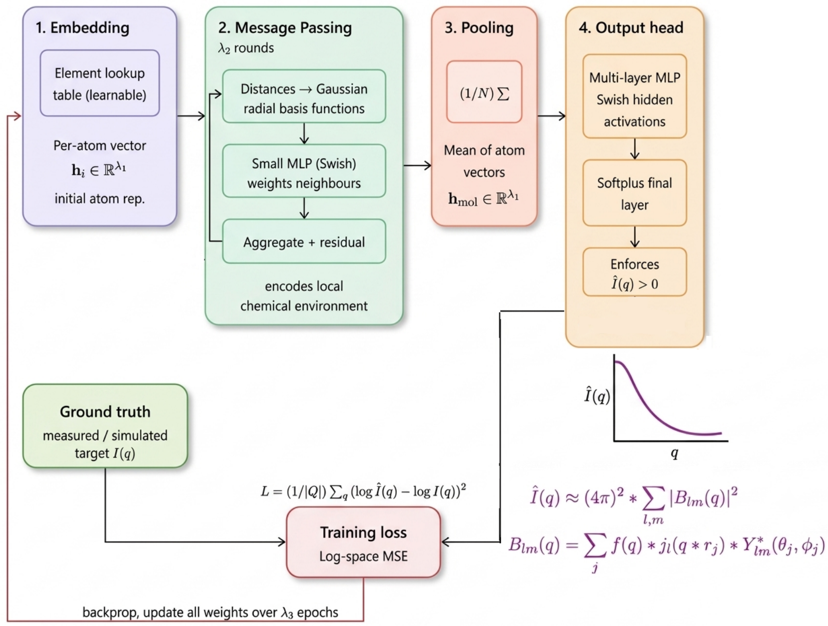
$$B_{lm}(q) = \sum_i f_i(q)\, j_l(qr_i)\, Y_{lm}^*(\theta_i, \phi_i), \qquad I(q) = (4\pi)^2 \sum_{l,m} |B_{lm}(q)|^2 \tag{2}$$

2 Data Processing

A dataset of **1,139,747 molecules** was assembled from 13 sources with diverse composition. In total, 474 unique atom/ion types are represented (charge-resolved, e.g. `O` and `O2-` are distinct). Full source attributions are provided in `sources.tsv`, embedded in the database. Please see for more information. Raw structures were obtained in various formats (.cif, .pdb, .xyz) and converted to a uniform Cartesian XYZ representation. Dummy and non-physical atom sites were removed, deuterium was mapped to hydrogen, and structures declaring zero atoms were discarded. Non-element placeholders (`X`, `M`, `Cn`, etc.) were excluded from the ionic makeup. $I(q)$ ground truths were then computed via the Stuhrmann decomposition (source code) at L=50 and stored in a compressed HDF5 database.

3 Proposed Architecture

The proposed network takes a molecule's atomic positions and element types as input and predicts $\hat{I}(q) \in \mathbb{R}^{|Q|}$. Let hyperparameters λ_1 be the hidden dimension, λ_2 the number of message-passing rounds, and λ_3 be the number of training epochs. **1) Embedding:** Atomic coordinate files are converted into a list of atoms, which are looked up in a trainable table, producing a vector $\mathbf{h}_i \in \mathbb{R}^{\lambda_1}$. This gives the network a learnable representation of each element. **2) Message passing (λ_2 rounds):** Each atom updates its vector by aggregating information from nearby neighbours. Interatomic



distances are encoded as Gaussian radial basis functions before being passed to a small MLP (with Swish activations) that weights each neighbour's contribution. A residual connection is added at each round. After λ_2 rounds, each atom's vector encodes its local chemical environment. **3) Pooling:** Atom vectors are averaged into a single fixed-size molecule vector $\mathbf{h}_{\text{mol}} \in \mathbb{R}^{\lambda_1}$, independent of molecule size. **4) Output head:** A multi-layer MLP with Swish hidden activations maps $\mathbf{h}_{\text{mol}}$ to $\hat{I}(q) \in \mathbb{R}^{|Q|}$. A Softplus activation on the final layer enforces strict positivity $\hat{I}(q) > 0$, consistent with the physical requirement that scattering intensity is always positive. **5) Backprop:** The model is trained a total of λ_3 epochs against ground truths. The network is trained to minimise a log-scale MSE loss, $\mathcal{L} = \frac{1}{|Q|}\sum_q \left(\log \hat{I}(q) - \log I(q)\right)^2$ which accounts for the orders-of-magnitude variation in intensity across the dataset.

4 Baseline Model

The baseline is a small MLP that takes as input only the atom-type histogram of a molecule (i.e. a count vector $\mathbf{c} \in \mathbb{Z}^{|\mathcal{E}|}$ recording how many atoms of each element type are present) and maps it directly to $\hat{I}(q) \in \mathbb{R}^{|Q|}$ via two hidden layers with Swish activations and a Softplus output. The model sees no atomic coordinates, no interatomic distances, and no graph structure whatsoever.

5 Ethical Considerations

All training data is open-source and fully attributed; no private or sensitive information is involved. The primary risk is misuse of model predictions: since the GNN is an approximation, predicted $I(q)$ curves could mislead structural biology or materials research if treated as exact without cross-validation against the ground-truth computation.

References

[1] K. T. Schütt, H. E. Sauceda, P.-J. Kindermans, A. Tkatchenko, and K.-R. Müller. SchNet – A deep learning architecture for molecules and materials. 148(24):241722, . ISSN 0021-9606, 1089-7690. doi: 10.1063/1.5019779.

[2] Johannes Gasteiger, Janek Groß, and Stephan Günnemann. Directional Message Passing for Molecular Graphs.

[3] Raghunathan Ramakrishnan, Pavlo O. Dral, Matthias Rupp, and O. Anatole Von Lilienfeld. Quantum chemistry structures and properties of 134 kilo molecules. 1(1):140022. ISSN 2052-4463. doi: 10.1038/sdata.2014.22.

[4] D. Svergun, C. Barberato, and M. H. J. Koch. *CRYSOL* – a Program to Evaluate X-ray Solution Scattering of Biological Macromolecules from Atomic Coordinates. 28(6):768–773. ISSN 0021-8898. doi: 10.1107/S0021889895007047.

[5] Kristof T. Schütt, Oliver T. Unke, and Michael Gastegger. Equivariant message passing for the prediction of tensorial properties and molecular spectra, .

[6] Luca Gelisio and Paolo Scardi. 100 years of Debye's scattering equation. 72(6):608–620. ISSN 2053-2733. doi: 10.1107/S2053273316014881.

[7] H. B. Stuhrmann. Interpretation of small-angle scattering functions of dilute solutions and gases. A representation of the structures related to a one-particle scattering function. 26(3):297–306. ISSN 0567-7394. doi: 10.1107/S0567739470000748.